Supplementary appendix to: Not All Embeddings Are Gaussian: Revisiting Adaptive HNSW Search: [Experiments & Analysis]

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A Additional results

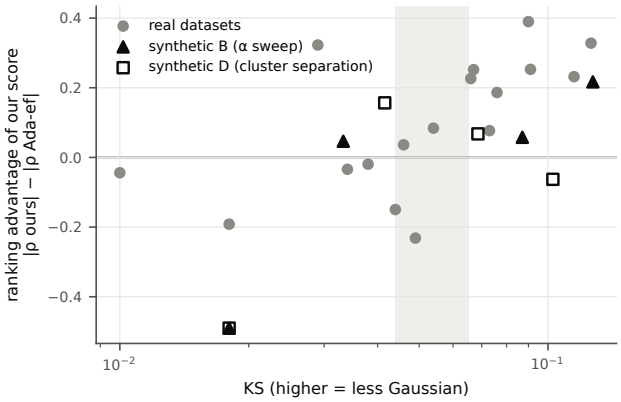


Figure A1: Ranking advantage of the empirical score against KS for the real datasets and for synthetic data with controlled anisotropy (B) and cluster separation (D).

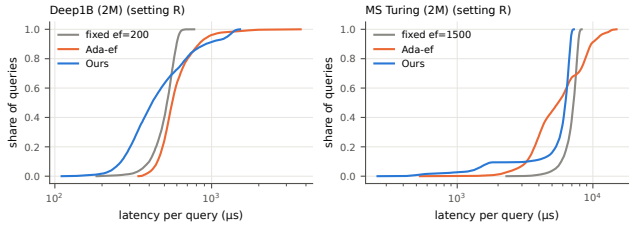


Figure A2: Per-query latency on Deep1B and MS Turing (setting R). The empirical score spends less time on easy queries and more on hard ones; a fixed *ef* spends about the same on all.

Table A1: Time per distance computation, relative to fixed-*ef* queries with the same distance count (setting P and R). Hard and easy: the 20% of queries PercEF gives the largest and smallest *ef*, timed under the fixed *ef* closest to PercEF's mean recall. Last column: PercEF's share of its distance computations spent on its hard queries.

dataset	setting	KS	hard / fixed-ef time at same DC	easy / fixed-ef time at same DC	PercEF / fixed-ef time at same DC	PercEF DC on hard queries (%)
MS Turing (2M)	P	0.010	1.002	0.998	1.026	95.1
MS Turing (2M)	R	0.010	1.000	0.996	0.998	97.9
GloVe-100	P	0.018	1.037	0.799	1.032	95.7
GloVe-100	R	0.018	1.037	0.799	1.021	97.3
BIGANN (2M)	P	0.029	1.035	0.863	1.114	50.5
BIGANN (2M)	R	0.029	1.044	0.855	1.069	39.0
LAION-I2I	P	0.034	1.185	0.841	1.067	58.0
LAION-I2I	R	0.034	1.184	0.820	1.112	54.3
DBpedia-1536	P	0.038	1.052	0.884	0.988	81.2
DBpedia-1536	R	0.038	1.059	0.839	1.023	66.1
MS MARCO-384	P	0.044	1.114	0.808	1.024	30.4
MS MARCO-384	R	0.044	1.133	0.824	1.016	30.8
COCO-I2I	P	0.046	1.121	0.943	1.068	43.8
COCO-I2I	R	0.046	1.133	0.928	1.066	35.9
Cohere-1024	P	0.049	1.072	0.849	1.076	63.7
Cohere-1024	R	0.049	1.087	0.865	1.103	71.2
Yahoo-MiniLM	P	0.054	1.083	0.890	1.006	44.9
Yahoo-MiniLM	R	0.054	1.085	0.879	0.990	39.8
COCO-T2I	P	0.056	1.006	1.006	1.048	100.0
COCO-T2I	R	0.056	0.991	0.991	0.990	100.0
ImageNet-ALIGN	P	0.058	1.013	1.013	1.048	100.0
ImageNet-ALIGN	R	0.058	0.989	0.989	1.029	100.0
DeepImage-96	P	0.066	1.069	0.916	1.076	52.0
DeepImage-96	R	0.066	1.087	0.912	1.075	43.6
Deep1B (2M)	P	0.067	1.092	0.891	1.048	34.4
Deep1B (2M)	R	0.067	1.083	0.883	1.065	43.9
Fashion-MNIST	P	0.073	1.093	0.979	1.017	30.3
Fashion-MNIST	R	0.073	1.094	0.979	1.012	25.6
Landmark-DINO	P	0.076	1.096	0.893	0.890	52.9
Landmark-DINO	R	0.076	1.066	0.881	0.890	49.0
Yambda	P	0.090	1.006	1.005	1.027	100.0
Yambda	R	0.090	0.999	0.978	1.004	100.0
GIST-960	P	0.091	1.006	1.004	0.984	98.8
GIST-960	R	0.091	1.004	1.001	0.987	99.4
iNat-ResNet	P	0.115	1.114	0.946	1.073	33.7
iNat-ResNet	R	0.115	1.121	0.916	1.025	30.0
SIFT-1M	P	0.126	1.051	0.897	1.002	42.7
SIFT-1M	R	0.126	1.059	0.893	1.002	28.9
Last.fm	P	0.216	0.998	0.998	1.046	100.0
Last.fm	R	0.216	0.996	0.996	1.049	100.0

Table A2: Tail-weighted alternatives to KS on the raw vectors. Tail *z*: distance of the empirical upper quantile from the Normal's, in standard deviations. Tail mass: $\log_{10}$ of the observed over the expected share beyond the Normal's 99.9% point. Margin: $|\rho_{\textbf{ours}}| - |\rho_{\textbf{Ada-ef}}|$, mean over P and R.

dataset	source	KS	AD/n	tail z 1e-2	tail z 1e-3	tail z 1e-4	tail mass 99.9%	rho margin
MS Turing (2M)	Big-ANN	0.0090	0.0001	-0.02	-0.02	+0.04	-0.11	-0.044
GloVe-100	ann-benchmarks	0.0167	0.0005	-0.04	+0.04	+0.12	-0.12	-0.192
BIGANN (2M)	Big-ANN	0.0299	0.0015	+0.08	-0.05	-0.26	-0.27	+0.323
LAION-I2I	LAION, shards 0-19	0.0328	0.0020	-0.09	+0.02	+0.06	-0.36	-0.034
DBpedia-1536	DBpedia, OpenAI embeddings	0.0387	0.0034	+0.31	+0.93	+1.49	+0.60	-0.019
MS MARCO-384	MS MARCO (MiniLM)	0.0444	0.0044	+0.57	+1.36	+2.29	+0.85	-0.150
Cohere-1024	Cohere, files 00-04	0.0452	0.0045	+0.52	+1.19	+1.91	+0.80	-0.232
COCO-I2I	ann-benchmarks	0.0458	0.0026	+0.51	+0.76	+0.75	+0.69	+0.036
Yahoo-MiniLM	VIBE	0.0543	0.0064	+0.67	+1.43	+2.12	+0.90	+0.084
COCO-T2I	ann-benchmarks	0.0563	0.0067	+0.75	+1.05	+1.09	+0.94	+0.035
ImageNet-ALIGN	VIBE	0.0566	0.0062	+0.77	+2.20	+3.18	+0.98	+0.054
DeepImage-96	ann-benchmarks	0.0656	0.0085	+0.57	+0.76	+0.75	+0.69	+0.226
Deep1B (2M)	Big-ANN	0.0674	0.0075	+0.61	+0.77	+0.75	+0.69	+0.252
Fashion-MNIST	ann-benchmarks	0.0711	0.0074	-0.38	-0.89	-1.38	-1.88	+0.077
Landmark-DINO	VIBE	0.0750	0.0121	+0.66	+1.09	+1.31	+0.81	+0.186
Yambda	Yambda	0.0902	0.0174	+1.03	+1.54	+1.50	+1.10	+0.390
GIST-960	ann-benchmarks	0.0913	0.0213	-0.78	-1.24	-1.64	-2.55	+0.253
iNat-ResNet	VIBE	0.1128	0.0248	+0.94	+1.70	+2.02	+0.74	+0.232
SIFT-1M	ann-benchmarks	0.1278	0.0301	-0.48	-0.97	-1.43	-3.13	+0.328
Last.fm	ann-benchmarks	0.2150	0.0918	+1.64	+7.25	+11.05	+1.17	+0.162

Table A3: Offline cost, round-2 datasets.

dataset	setting	Ada-ef statistics (s)	Ada-ef table (s)	Ada-ef memory (B)	ours bins (s)	ours min-ef (s)	ours memory (B)
MS Turing (2M)	P	5.2	5.1	41,320	0.7	7.4	824
MS Turing (2M)	R	5.2	206.2	41,528	0.7	257.5	768
GloVe-100	P	–	–	41,496	–	–	672
GloVe-100	R	–	–	41,608	–	–	820
BIGANN (2M)	P	4.8	0.3	66,600	0.8	0.3	936
BIGANN (2M)	R	4.8	3.0	66,648	0.8	3.3	936
LAION-I2I	P	–	–	1,053,000	–	–	2,472
LAION-I2I	R	–	–	1,053,448	–	–	2,472
DBpedia-1536	P	–	–	9,450,048	–	–	6,540
DBpedia-1536	R	–	–	9,450,232	–	–	6,540
MS MARCO-384	P	–	–	593,560	–	–	1,952
MS MARCO-384	R	–	–	593,696	–	–	1,960
COCO-I2I	P	1.7	0.1	1,053,160	0.2	0.1	2,472
COCO-I2I	R	1.7	1.0	1,053,472	0.2	1.1	2,472
Cohere-1024	P	–	–	4,203,160	–	–	4,520
Cohere-1024	R	–	–	4,203,248	–	–	4,476
Yahoo-MiniLM	P	–	–	593,552	–	–	1,912
Yahoo-MiniLM	R	–	–	593,552	–	–	1,836
COCO-T2I	P	1.7	0.1	1,053,176	0.2	0.1	2,472
COCO-T2I	R	1.7	8.0	1,052,680	0.2	9.7	2,072
ImageNet-ALIGN	P	–	–	1,644,088	–	–	2,984
ImageNet-ALIGN	R	–	–	1,643,528	–	–	2,584
DeepImage-96	P	–	–	37,728	3.1	–	808
DeepImage-96	R	–	–	37,904	3.1	–	808
Deep1B (2M)	P	5.8	0.2	37,800	0.6	0.2	808
Deep1B (2M)	R	5.8	3.0	38,072	0.6	3.3	808
Fashion-MNIST	P	–	–	2,464,976	–	–	3,560
Fashion-MNIST	R	–	–	2,465,144	–	–	3,560
Landmark-DINO	P	–	–	2,366,032	–	–	3,496
Landmark-DINO	R	–	–	2,366,096	–	–	3,496
Yambda	P	–	–	66,568	–	–	936
Yambda	R	–	–	66,600	–	–	936
GIST-960	P	–	–	3,694,168	–	–	3,888
GIST-960	R	–	–	3,694,160	–	–	3,888
iNat-ResNet	P	–	–	16,793,952	–	–	8,616
iNat-ResNet	R	–	–	16,794,120	–	–	8,616
SIFT-1M	P	–	–	66,568	0.4	–	936
SIFT-1M	R	–	–	66,600	0.4	–	936
Last.fm	P	0.8	0.4	17,436	0.1	2.1	684
Last.fm	R	0.8	0.3	17,428	0.1	0.4	284